

# Predicting Growth and Remodeling of Cardiovascular Tissue

change in mass

change in microstructure

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With much inspiration from Drs. Amadeus Gebauer, Jonas Eichinger, and Jay Humphrey

Teacher Version

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# The Origin of Mechanobiology

**Davis' law, Henry Gassett Davis** (1807–1896, USA), 1867:

*“Ligaments, or any soft tissue, when put under even a moderate degree of tension, if that tension is unremitting, will elongate by the addition of new material.”*

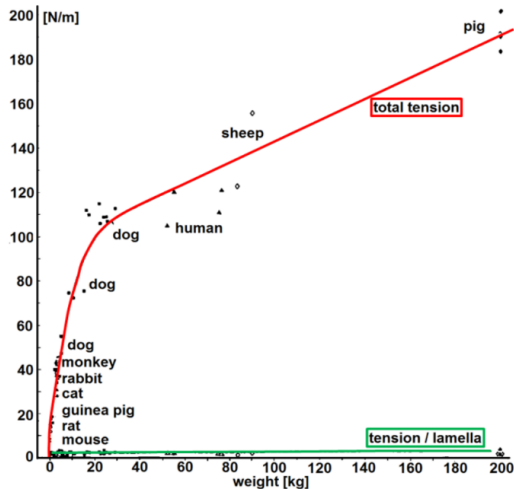
**Wolff's law, Julius Wolff** (1836–1902, Germany), 1892 (*Das Gesetz der Transformation der Knochen*, the law of bone transformation):

*“im Gefolge primärer Abänderungen der Form und Inanspruchnahme oder auch bloß der Inanspruchnahme der Knochen [sich] bestimmte, nach mathematischen Regeln eintretende Umwandlungen der inneren Architektur und ebenso bestimmte, denselben mathematischen Regeln folgende sekundäre Umwandlungen der äußeren Form der betreffenden Knochen vollziehen.”*

- bone mass proportional to load (higher / lower load  $\Rightarrow$  growth / atrophy)
- first *mathematical* relation between load and growth & remodeling in biological tissue

# A Brief Introduction to Mechanobiology of Blood Vessels

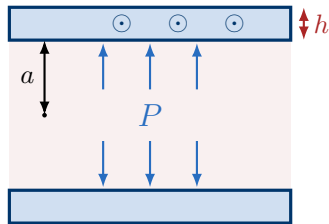
# Vascular tissue has a preferred mechanical state



Tension  $T$  varies by a **factor of 26**

$$T = P a$$

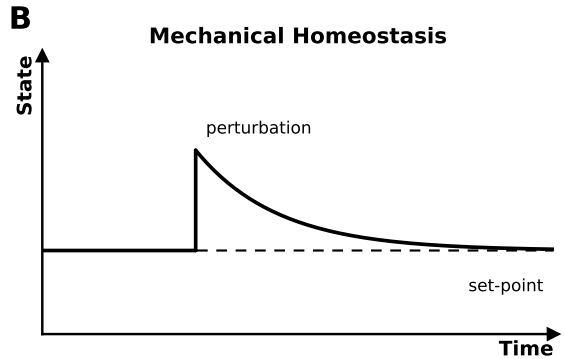
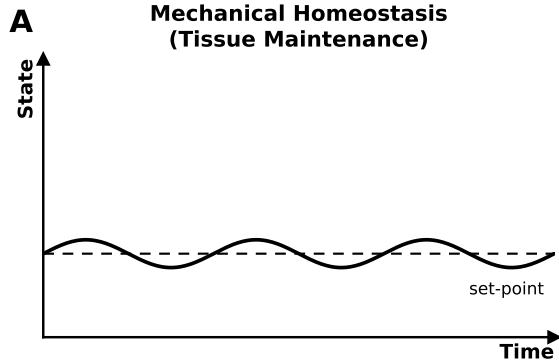
$T$  and  $\sigma_\theta$



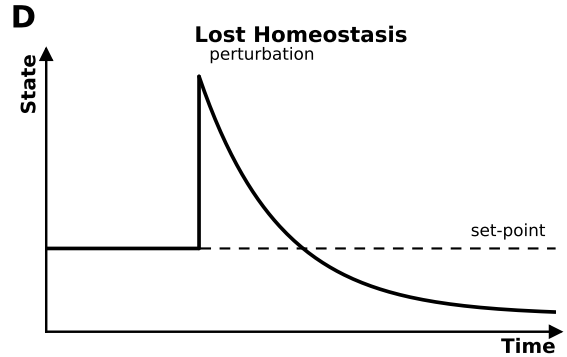
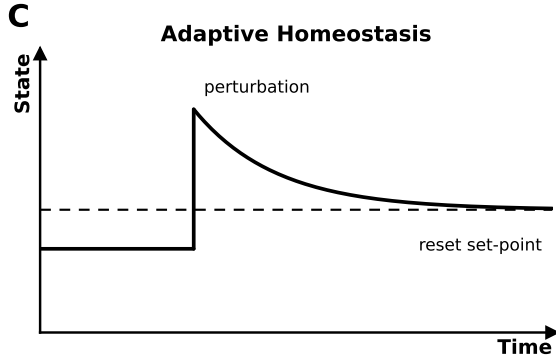
Tension / lamella varies by a **factor of 2.8**

$$\frac{T}{h} = \frac{P a}{h} = \sigma_\theta$$

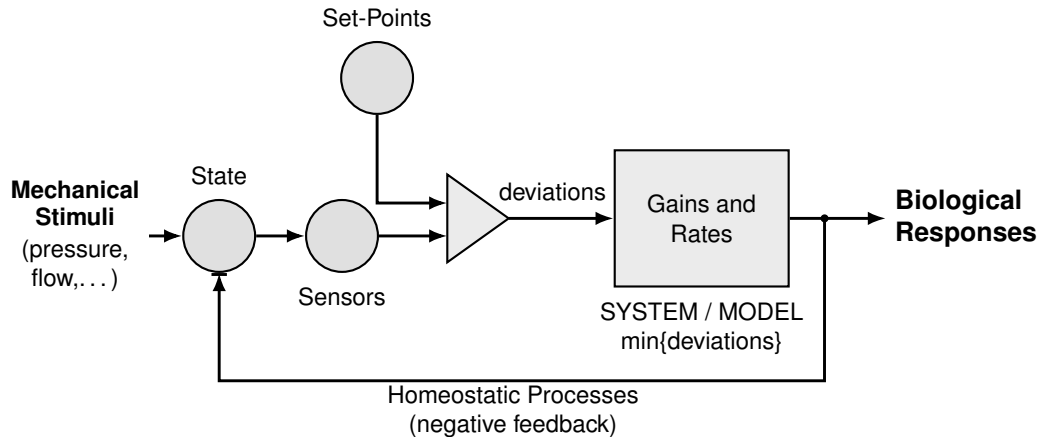
# Mechanical Homeostasis



# (Mal)adaptation

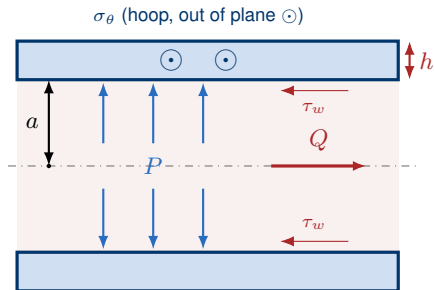


# Mechanobiological Regulation



# An Idealized Blood Vessel

A vessel is a **pressurised, perfused tube**. Two loads act on the wall: **blood pressure**  $P$  pushes it outward, and **blood flow**  $Q$  drags its inner surface. Cells sense both.



$a$  inner radius,  $h$  wall thickness

$P$  lumen pressure  $\rightarrow$  **circumferential stress**  $\sigma_\theta \sim 100$  kPa

$Q$  volumetric flow  $\rightarrow$  **wall shear stress**  $\tau_w \sim 1$  Pa



# Two Loads, Two Formulae, Two Adaptations

Pressure  $P$  sets the **hoop stress**; flow  $Q$  sets the **wall shear stress**. Each has a set-point; rearranging shows the geometry that restores it.

**Pressure** (Laplace)

$$\sigma_{\theta} = \frac{Pa}{h} \quad (1)$$

**Flow** (Poiseuille)

$$\tau_w = \frac{4\mu Q}{\pi a^3} \quad (2)$$

## ✂ Your turn: Pressure adaptation

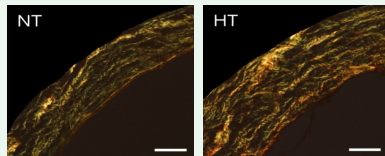
**Question.** Pressure **doubles**. How does the wall thickness change?  $\sigma_\theta = \frac{Pa}{h}$

## ✓ Answer: Pressure adaptation

$P'/h' = P/h$  with  $P' = 2P$  gives

$$h' = \left(\frac{P'}{P}\right) h = 2h. \quad (3)$$

Wall thickens *in proportion* to pressure.



$P \uparrow$

## ✂ Your turn: Flow adaptation

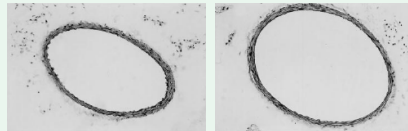
**Question.** Flow rises **8-fold**. How does the radius change?  $\tau_w = \frac{4\mu Q}{\pi a^3}$

## ✓ Answer: Flow adaptation

$(a'/a)^3 = Q'/Q = 8$ , so

$$a' = \left(\frac{Q'}{Q}\right)^{1/3} a = 2a. \quad (4)$$

**Reading.** Radius grows as the *cube root* of flow.



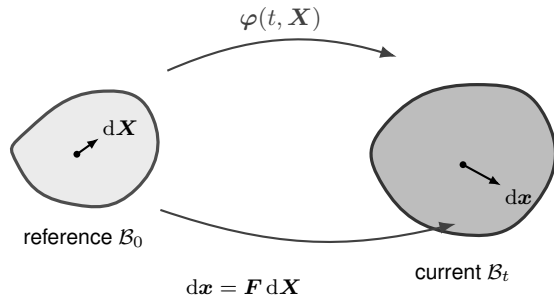
$Q \uparrow$

# Finite-Strain Fundamentals

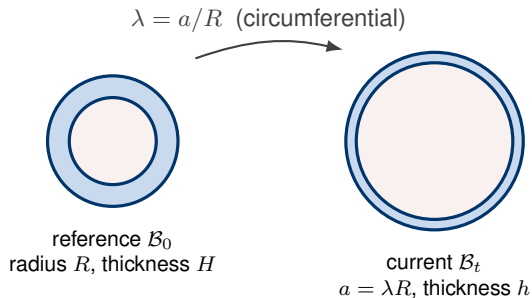
# Reference and Current Configuration

A body is described by a **reference**  $\mathcal{B}_0$  (points  $\mathbf{X}$ ) and a **current**  $\mathcal{B}_t$  (points  $\mathbf{x}$ ) configuration, related by the motion  $\mathbf{x} = \varphi(t, \mathbf{X})$ .

## 3D 3D geometric model



## 0D 0D reduced model



# The Deformation Gradient

The **deformation gradient** maps reference line elements to current ones.

**3D** a  $3 \times 3$  tensor

$$d\mathbf{x} = \mathbf{F} d\mathbf{X}, \quad \mathbf{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \mathbf{X}}, \quad J = \det \mathbf{F} \quad (5)$$

$J$  is the volume ratio, **incompressibility** means  $J = 1$ .

**0D** a scalar

$$\mathbf{F} \longrightarrow \lambda = \frac{\text{current length}}{\text{reference length}}, \quad \lambda = 1 \text{ undeformed.}$$

# Strain: A Rotation-Free Measure of Stretch

Removing rigid rotation gives the **right Cauchy–Green** tensor and the **Green–Lagrange strain**.

**3D** symmetric  $3 \times 3$  tensors

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = (\mathbf{R}\mathbf{U})^T \mathbf{R}\mathbf{U} = \mathbf{U}^T \mathbf{R}^T \mathbf{R} \mathbf{U} = \mathbf{U}^2, \quad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}) \quad (6)$$

**0D** scalars

$$C = \lambda^2, \quad E = \frac{1}{2}(\lambda^2 - 1).$$

# Stress: "Force per Area"

**Cauchy stress:** *current* force per unit *current* area.

**3D** spatial symmetric  $3 \times 3$  tensor (from 2nd Piola–Kirchhoff stress  $S$ )

$$\boldsymbol{\sigma} = \frac{1}{J} \mathbf{F} \mathbf{S} \mathbf{F}^T, \quad \mathbf{S} = 2 \frac{\partial W}{\partial \mathbf{C}}, \quad W \text{ strain energy density function.}$$

**0D** scalar reduction ( $C = \lambda^2$ ,  $J = 1$ ), the  $\sigma_\theta$  in Laplace (1)

$$\sigma = \frac{1}{J} \lambda S \lambda = \lambda^2 S = 2\lambda^2 \frac{\partial W}{\partial C} = \lambda \frac{dW}{d\lambda}.$$



# Hyperelasticity: Stress from a Stored-Energy Function

A **hyperelastic** material stores energy  $W$  (deformation is path-independent).

**3D** energy  $W(\mathbf{C})$ , stress  $\mathbf{S} = 2 \partial W / \partial \mathbf{C}$

$$W_{\text{neo-Hooke}} = \frac{c}{2} (I_1 - 3), \quad W_{\text{Fung}} = \frac{c_1}{4c_2} \left[ e^{c_2(I_1-1)^2} - 1 \right]$$

Invariants of  $\mathbf{C}$ :  $I_1 = \text{tr } \mathbf{C}$  (isotropic, elastin);  $I_4 = \mathbf{M} \cdot \mathbf{C} \mathbf{M} = \lambda_{\text{fib}}^2$ , the squared stretch along the reference fiber direction  $\mathbf{M}$  (anisotropy, collagen).

**0D** reduction used here, one stretch  $\lambda$ , scalar  $\sigma = \lambda \, dW/d\lambda$

$$\sigma_{\text{neo-Hooke}} = c (\lambda^2 - \lambda^{-1}), \quad \sigma_{\text{Fung}} = c_1 \lambda^2 (\lambda^2 - 1) e^{c_2(\lambda^2-1)^2} \quad (7)$$

# Kinematic Growth

# Multiplicative Decomposition: Growth vs. Elasticity

Split the deformation into a stress-free **growth** part  $F_g$  and a stress-carrying **elastic** part  $F_e$ .

**3D** a product of tensors

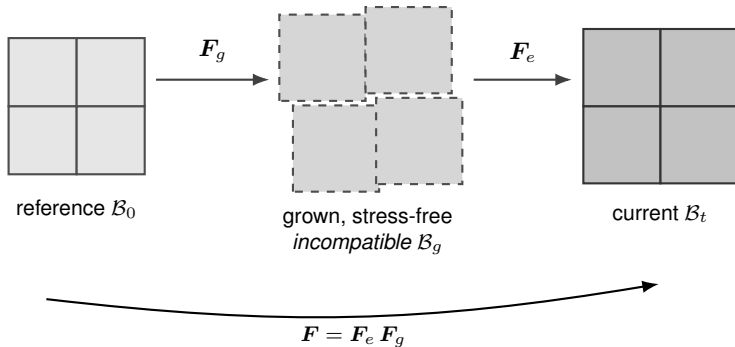
$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g, \quad \text{only } \mathbf{F}_e \text{ stores energy.}$$

**0D** scalar stretches

$$\lambda = \lambda_e \lambda_g \tag{8}$$

# Multiplicative Decomposition: Growth vs. Elasticity

$F_g$  yields an *incompatible* stress-free state,  $F_e$  restores compability via residual stress.



# Kinematic Growth Tensor

$\mathbf{F}_g$  encodes *how* the tissue is assumed to add mass. A common choice grows the tissue an amount  $\theta$  along a fixed direction  $\mathbf{f}$ :

**3D** prescribed growth tensor

$$\mathbf{F}_g = \mathbf{I} + (\theta - 1) \mathbf{f} \otimes \mathbf{f} = \text{diag}(1, 1, \theta) \quad (9)$$

**0D** growth stretch

$$\lambda_g \text{ (scalar growth stretch),} \quad \theta = \det \mathbf{F}_g = M/M_0.$$

## Growth Law

The growth *pattern*  $F_g$  is typically fixed. Its rate  $\dot{F}_g = \dot{\theta} \mathbf{f} \otimes \mathbf{f}$  reduces to *one* scalar ODE:

$$\frac{d\theta}{dt} = k_g \theta \left( \frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right) \quad (10)$$

Since  $k_g \theta > 0$ , the sign of the response follows the stress deviation:

$$\frac{d\theta}{dt} \begin{cases} > 0 & \bar{\sigma} > \bar{\sigma}_h & \text{(growth)} \\ = 0 & \bar{\sigma} = \bar{\sigma}_h & \text{(homeostasis)} \\ < 0 & \bar{\sigma} < \bar{\sigma}_h & \text{(atrophy)} \end{cases}$$

# Getting Started with the Code

One runner drives *every* exercise: you never edit Python, just point it at a small input file. **First-time setup** (needs Python  $\geq 3.10$ ; dependencies: NumPy, SciPy, Matplotlib, PyYAML):

```
git clone https://github.com/YaleCBL-Teaching/Growth-Remodeling
cd Growth-Remodeling
```

**uv** (recommended):  
uv sync

**pip** (your own Python):  
python -m venv .venv  
pip install -e .

## What is where:

- `run.py`: the single runner for all exercises
- `configs/`: one YAML input file per exercise (*you edit these*)
- `src/gr/`: the models & maths (read, don't edit)
- `docs/`: documentation
- `slides/`: semianr notes

## ⚙️ Setup: Example (hypertensive artery, ex01)

**Example.** A pressurised artery is made *hypertensive* by a sustained pressure step. Kinematic growth thickens the wall. Does the wall stress  $\bar{\sigma}/\bar{\sigma}_h$  return to the *prescribed* set-point?



# Running an Exercise

Point the runner at the input file (`uv run`, or plain `python` once the `venv` is activated):

```
uv run python run.py configs/ex01_kinematic_growth.yaml
```

**You get:** the key numbers ( $\bar{\sigma}/\bar{\sigma}_h$ , mass, diverged) print to the terminal, and the time history opens as a plot.

**Change one value:** edit the YAML, or override on the command line:

```
uv run python run.py configs/ex01_kinematic_growth.yaml -set insult.pressure_factor=2
```

**Parametric study:** one curve per value:

```
... -sweep insult.pressure_factor=1.5,2,3
```

## Your turn: Kinematic growth

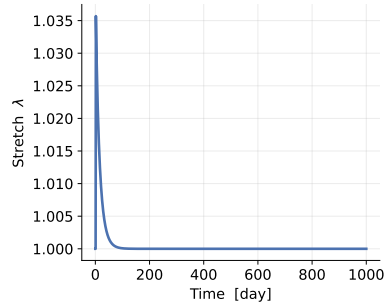
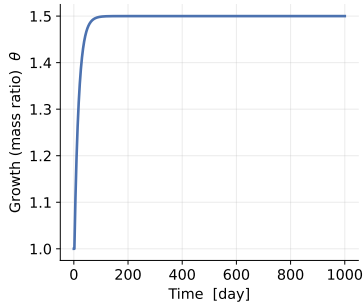
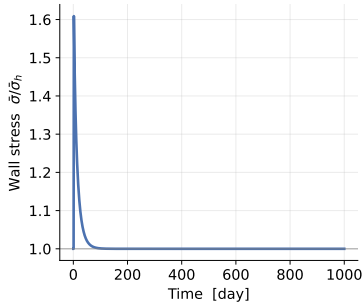
**Question.** Does the wall stress return to the prescribed set-point? Try larger pressure steps. Does the outcome ever change?

**Hints.**  $\bar{\sigma}/\bar{\sigma}_h$  settles at 1 while  $\theta$  climbs to the pressure factor. Under pure hypertension it *always* adapts. Runaway needs the mixture / aneurysm story (later).

## ✓ Answer: Kinematic growth

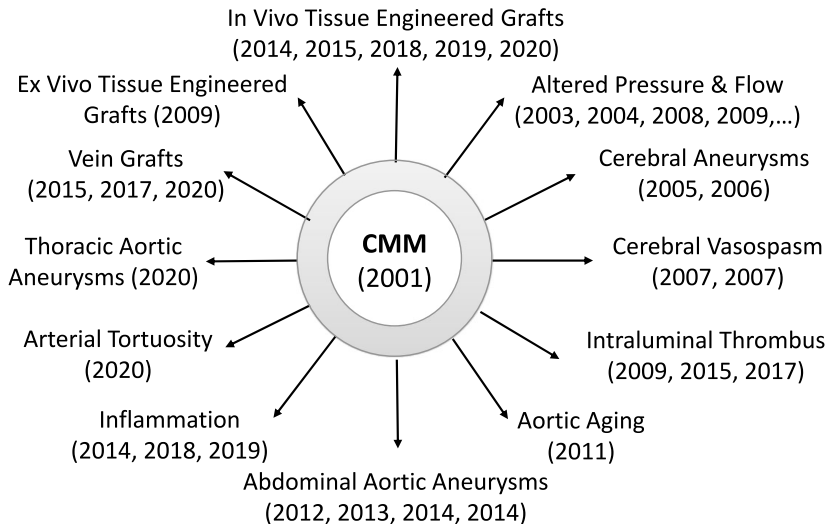
The stress plateau *always* lands on the prescribed set-point. The wall simply thickens more for a larger pressure step. Kinematic growth *imposes* stability; it cannot predict runaway.

Kinematic growth of the artery under hypertension (P: 1.0  $\rightarrow$  1.5x)



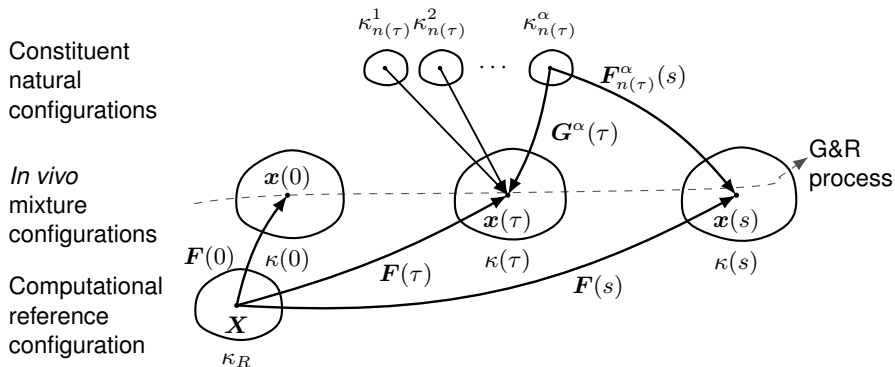
# The Constrained Mixture Model

# One Framework, Many Applications



**Ubiquitous Vascular G&R**

# The Idea: Track Constituents Over Time



$\kappa$ : configuration •  $\tau$ : deposition time •  $s$ : current time •  $\alpha$ : constituent •  $G^\alpha$ : deposition stretch •  $F_{n(\tau)}^\alpha$ : constituent deformation

# Load-Bearing Vessel Constituents

The wall is a mixture of three constituents, each with a reference **mass fraction**  $\phi_0^\alpha$  ( $\sum_\alpha \phi_0^\alpha = 1$ ) and a **deposition stretch**  $G^\alpha$ , the prestretch cells lay it down at, which *defines* its homeostatic stress  $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$ ; the mixture set-point is  $\bar{\sigma}_h = \sum_\alpha \phi_0^\alpha \sigma_h^\alpha$ .

| Constituent          | Role                         | $\phi_0$ | $G^\alpha$ | law; turnover $1/k_d$            |
|----------------------|------------------------------|----------|------------|----------------------------------|
| <b>Elastin</b>       | bears load at low stretch    | 0.30     | 1.40       | neo-Hooke; <b>none</b> (decades) |
| <b>Collagen</b>      | stiff, crimped; anti-rupture | 0.35     | 1.08       | Fung; fast ( $\sim 80$ d)        |
| <b>Smooth muscle</b> | passive + active; the sensor | 0.35     | 1.20       | Fung; fast ( $\sim 80$ d)        |

**Elastin cannot be renewed** ( $k_d^e = 0$ ), its loss drives *aneurysm*; collagen & SMC **turn over**, so the wall can *adapt*.

# Every Cohort Shares One Stretch

A cohort of  $\alpha$  deposited at  $\tau$  is laid down at  $G^\alpha$  and then stretches *with the mixture*: every constituent and cohort is **constrained** to the same tissue stretch  $\lambda(s)$ .

**3D** constituent-specific deformation

$$\mathbf{F}_{n(\tau)}^\alpha(s) = \mathbf{F}(s) \mathbf{F}^{-1}(\tau) \mathbf{G}^\alpha(\tau).$$

**0D** scalar elastic stretch

$$\lambda_{n(\tau)}^\alpha(s, \tau) = \frac{\lambda(s)}{\lambda(\tau)} G^\alpha \quad (11)$$



# Mass Balance

Track each constituent's **apparent mass density**  $\rho^\alpha$  per current mixture volume,  $\rho = \sum_\alpha \rho^\alpha$ .

Spatial mass balance (one per constituent):

$$\frac{\partial \rho^\alpha}{\partial s} + \underbrace{\operatorname{div}(\rho^\alpha \mathbf{v}^\alpha)}_{\text{transport}} = \underbrace{\bar{m}^\alpha}_{\text{net mass exchange}}, \quad \alpha = 1, \dots, N.$$

The net rate is *signed*:

$$\bar{m}^\alpha = \underbrace{m^\alpha}_{\text{production} \geq 0} - \underbrace{r^\alpha}_{\text{removal} \geq 0} \quad \left\{ \begin{array}{ll} < 0 & \text{atrophy} \\ = 0 & \text{maintenance} \\ > 0 & \text{growth} \end{array} \right.$$

# From Balance to a Heredity Integral

**Assumption 1 (constrained mixture):** every constituent moves with the mixture,  $v^\alpha = v$ .

**Assumption 2 (prescribed rates):** a true production rate  $m^\alpha(\tau) \geq 0$  and a survival fraction  $q^\alpha(s, \tau) \in [0, 1]$ . Integrating from the distant past to  $s$ :

$$\rho^\alpha(s) = \underbrace{\rho^\alpha(0) Q^\alpha(s)}_{\text{survivors of original material}} + \underbrace{\int_0^s m^\alpha(\tau) q^\alpha(s, \tau) d\tau}_{\text{deposited at } \tau, \text{ still present at } s} \quad (12)$$

with  $Q^\alpha(0) = 1$ . **Assumption 3:** bulk density is nearly constant,  $\rho(s) = \rho(0)$ , mass changes track volume changes.

# Linear Momentum Balance

A rule-of-mixtures for the stresses (no body forces) satisfies the classical linear-momentum balance:

$$\operatorname{div} \boldsymbol{\sigma} = \rho \boldsymbol{a} \xrightarrow{\text{quasi-static}} \operatorname{div} \boldsymbol{\sigma} = \mathbf{0}.$$

**Quasi-static assumption:** G&R evolves slowly at every time  $s$ , so the inertia  $\rho \boldsymbol{a}$  is negligible.

# Constitutive Relations: Cauchy Stress

In finite elasticity the Cauchy stress follows from a stored-energy function  $W$  (per unit reference mixture volume):

$$\boldsymbol{\sigma}(s) = \frac{2}{J} \mathbf{F}(s) \frac{\partial W(s)}{\partial \mathbf{C}(s)} \mathbf{F}^T(s), \quad \mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad J = \det \mathbf{F}.$$

Rule of mixtures:  $W = \sum_{\alpha} \hat{W}^{\alpha}$ . Closing the model needs **three constituent-specific relations**:

- a *stored energy*  $W^{\alpha}$ ,
- a *production rate*  $m^{\alpha}$ , and
- a *removal* (survival)  $q^{\alpha}$ .

# Three Constituent-Specific Constitutive Relations

1. **Material:** constituent stored energy  $W^\alpha$  (e.g., neo-Hookean, Fung-type)
2. **Rate of production:** up with wall stress and inflammation, down with wall shear:

$$m^\alpha(\tau) = m_o^\alpha \left( 1 + \underbrace{\frac{K_\sigma^\alpha \Delta\sigma}{\text{circumferential stress}}}_{\text{circumferential stress}} - \underbrace{\frac{K_{\tau_w}^\alpha \Delta\tau_w}{\text{wall shear stress}}}_{\text{wall shear stress}} + \underbrace{\frac{K_\varphi^\alpha \Delta\rho_\varphi}{\text{inflammation}}}_{\text{inflammation}} \right) \geq 0 \quad (13)$$

3. **Rate of removal:** a survival fraction that decays faster under stress perturbation:

$$q^\alpha(s, \tau) = \exp\left(-\int_\tau^s k_d^\alpha (1 + \omega \Delta\sigma^2(t) + \dots) dt\right) \in [0, 1] \quad (14)$$

# Constituent Stored-Energy Functions

Each cohort carries its *own* stored energy  $W^\alpha$ . Summing over cohorts with the *same* heredity structure gives the apparent (mixture-level) energy:

$$\rho \hat{W}^\alpha(s) = \underbrace{\rho^\alpha(0) Q^\alpha(s) W^\alpha(\mathbf{F}_{n(0)}^\alpha(s))}_{\text{original material}} + \underbrace{\int_0^s m^\alpha(\tau) q^\alpha(s, \tau) W^\alpha(\mathbf{F}_{n(\tau)}^\alpha(s)) d\tau}_{\text{later cohorts}} \quad (15)$$

, with elastic deformation

$$\mathbf{F}_{n(\tau)}^\alpha(s) = \mathbf{F}(s) \mathbf{F}^{-1}(\tau) \mathbf{G}^\alpha(\tau) \quad (16)$$

. The mixture energy  $W = \sum_\alpha \hat{W}^\alpha$  closes the mixture stress.

## Special Cases: Before Perturbation and Homeostasis

(I) At  $s = 0$  (just before a perturbation) only original material is present, and  $Q^\alpha(0) = 1$ :

$$\rho \hat{W}^\alpha(0) = \rho^\alpha(0) W^\alpha(\mathbf{F}_{n(0)}^\alpha(0)) \Rightarrow \hat{W}^\alpha(0) = \underbrace{\frac{\rho^\alpha(0)}{\rho(0)}}_{\phi^\alpha(0)} W^\alpha(0),$$

recovering the rule of mixtures  $W(0) = \sum_\alpha \phi^\alpha(0) W^\alpha(0)$ .

(II) **Homeostasis:** turnover in an *unchanging* configuration, so  $\mathbf{F}_{n(\tau)}^\alpha(s) = \mathbf{F}_{n(0)}^\alpha(s)$  (independent of  $\tau$ ). Then  $W^\alpha$  factors out and, using the mass balance,

$$\rho \hat{W}^\alpha(s) = \underbrace{\left( \rho^\alpha(0) Q^\alpha(s) + \int_0^s m^\alpha q^\alpha d\tau \right)}_{= \rho^\alpha(s)} W^\alpha(\mathbf{F}_{n(0)}^\alpha(s)) \Rightarrow \hat{W}^\alpha(s) = \phi^\alpha(s) W^\alpha(s). \quad (17)$$

# Perfect Maintenance

Take *constant* rates:  $m^\alpha = m_0$ ,  $Q^\alpha(s) = e^{-K^\alpha s}$ ,  $q^\alpha(s, \tau) = e^{-K^\alpha(s-\tau)}$ . The mass balance integrates to

$$\rho^\alpha(s) = \underbrace{\rho^\alpha(0) e^{-K^\alpha s}}_{\text{original decays}} + \underbrace{\frac{m_0}{K^\alpha} (1 - e^{-K^\alpha s})}_{\text{production accumulates}}.$$

A *time-invariant* density ( $\rho^\alpha(s) = \rho^\alpha(0)$  for all  $s$ ) requires the two to balance exactly:

$$\text{perfect maintenance} \iff \frac{m_0}{K^\alpha} = \rho^\alpha(0). \quad (18)$$

Production  $m_0$  then exactly offsets removal  $K^\alpha \rho^\alpha$  (continuous turnover, unchanging tissue).



## ✂ Your turn: Computational Cost?

**Question.** At each of  $N$  time steps the mixture stress sums over *all* cohorts deposited so far. How does the total cost scale with  $N$ ?

**Hints.** Step  $k$  re-sums  $\sim k$  stored cohorts; add up  $\sum_{k=1}^N k$ .

## ✓ Answer: Computational Cost?

Step  $k$  re-sums the  $\sim k$  cohorts stored so far, so the total work is

$$\sum_{k=1}^N k = \frac{N(N+1)}{2} = \mathcal{O}(N^2)$$

**quadratic** in the number of time steps. The *homogenized* model (next) removes this history sum, cutting the cost to  $\mathcal{O}(N)$ .

## ⚙️ Setup: Example 2: Turnover under Hypertension (ex02)

**Example.** The same artery, now with collagen & SMC turning over (elastin does not). Which constituent grows to carry the load, and does turnover restore the set-point? Equilibrium each step:  $\bar{\sigma}(\lambda) = \bar{\sigma}_h \gamma \lambda^2 / m$  (load factor  $\gamma$ , mass ratio  $m$ ).

### Run it.

```
uv run python run.py configs/ex02_constrained_mixture.yaml
```

Sets each constituent's `turnover_days` ( $1/k_d$ ), gain ( $K_\sigma$ ), and the `insult`, hypertension ( $\gamma$ ) or aneurysm (`elastin_surviving`).

### Parametric study.

```
... -set model.constituents.collagen.gain=3.0
```

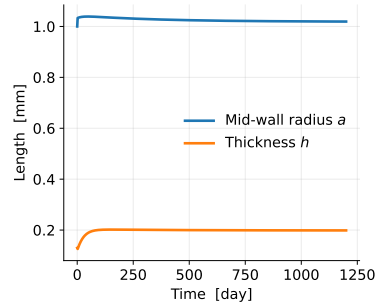
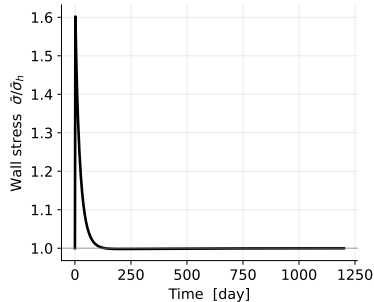
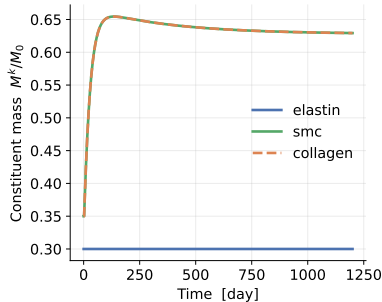
## Your turn: Constrained Mixture Model

**Question.** Does the same tissue *adapt* to hypertension but *run away* on the aneurysm insult? Which constituent grows to carry the load elastin used to bear?

## ✓ Answer: Constrained Mixture Model

Mild hypertension **adapts**: collagen and SMC grow, tissue stress returns to homeostatic. A severe elastin insult can **run away**. Faster turnover or higher gain reach the new state sooner.

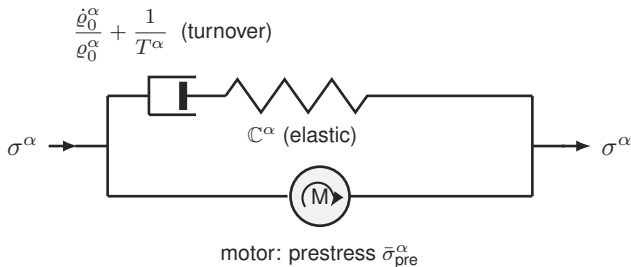
Full constrained-mixture model: adaptation to hypertension (artery)



# Homogenized CMM

# The Mechanical Analog: A Maxwell Fluid with a Motor

Each constituent behaves like a **Maxwell element in parallel with a motor**: turnover relaxes stress toward a *prestress*, not zero.



|                                                          |                                  |
|----------------------------------------------------------|----------------------------------|
| $\sigma^\alpha$                                          | constituent stress (the load)    |
| $\mathbb{C}^\alpha$                                      | elastic stiffness (spring)       |
| $\dot{\varrho}_0^\alpha / \varrho_0^\alpha + 1/T^\alpha$ | turnover + degradation (dashpot) |
| $\bar{\sigma}_{\text{pre}}^\alpha$                       | deposition prestress (motor)     |

## Back to the full CMM (0D):

- spring  $\leftrightarrow$  constituent law  $\sigma^\alpha(\lambda_e^\alpha)$
- dashpot  $\leftrightarrow$  mass turnover / removal  $k_d^\alpha$
- motor  $\leftrightarrow$  deposition stretch  $G^\alpha$

# Growth (Mass) and Remodeling (Natural Configuration)

Two 0D ODEs per constituent replace the cohort history.

**Growth:** mass production  $\propto$  mass, driven by the tissue-stress deviation.

$$\frac{dM^\alpha}{dt} = M^\alpha K_\sigma^\alpha k_d^\alpha \left( \frac{\bar{\sigma}}{\bar{\sigma}_h} - 1 \right) \quad (19)$$

**Remodeling:** new mass enters at natural stretch  $\lambda/G^\alpha$ , old mass leaves at the current  $\lambda_n^\alpha$  (elastic stretch  $\lambda_e^\alpha = \lambda/\lambda_n^\alpha$ ).

$$\frac{d\lambda_n^\alpha}{dt} = \frac{m^\alpha}{M^\alpha} \left( \frac{\lambda}{G^\alpha} - \lambda_n^\alpha \right) \quad (20)$$

with production  $m^\alpha = k_d^\alpha M^\alpha \Upsilon^\alpha$  and stimulus  $\Upsilon^\alpha = 1 + K_\sigma^\alpha (\bar{\sigma}/\bar{\sigma}_h - 1)$ . Holding  $\lambda$  fixed, remodeling relaxes each constituent's stress to its prestress  $\sigma_h^\alpha$  (time constant  $\sim 1/k_d^\alpha$ ), the Maxwell-with-motor behaviour.

## ✂ Your turn: Why Is the Homogenized Model Cheap?

**Question.** How does the cost of the homogenized constrained mixture model scale with the number of steps  $N$ , compared with the full model?

## ✓ Answer: Why Is the Homogenized Model Cheap?

The state is *fixed-size*, two numbers  $(M^\alpha, \lambda_n^\alpha)$  per constituent, and each step just advances two ODEs, with *no* history to re-sum:

$$\text{homogenized: } \mathcal{O}(N) \quad \text{vs.} \quad \text{full CMM: } \mathcal{O}(N^2)$$

Same time trajectory, far cheaper, and the fixed state maps onto a viscoelastic (Maxwell-with-motor) material, so it drops straight into standard finite-element solvers.



## ✂ Your turn: When do homogenized and full agree?

**Question.** The homogenized model keeps only the *mean* natural configuration per constituent. Under what assumptions should it match the full deposition-history model and when would you expect them to diverge?

## ✓ Answer: When do homogenized and full agree?

They agree when each constituent's cohorts share *nearly the same* natural configuration i.e. slow, gradual loading where deposition stretch  $G^{\alpha}$  and turnover are roughly stationary, so the cohort spread stays narrow and the mean represents it well. **They diverge** under fast or strongly time-varying insults, when cohorts deposited at very different states coexist and a single mean stretch can no longer capture the distribution.

## ⚙️ Setup: Example (homogenized vs. full, ex03)

**Example.** Run both mixture theories on the *same* artery and insult. How closely does the homogenized model track the full model, and how much faster is it?

### Run it.

```
uv run python run.py configs/ex03_homogenized_vs_full.yaml
```

Here study: compare overlays constrained\_mixture, homogenized\_cmm and kinematic\_growth on one scenario, and prints each runtime and the mass-ratio gap between the first two (full vs. homogenized).

**Parametric study.** Stress the comparison with a harsher, longer insult:

```
... -set insult.elastin_surviving=0.3 -set insult.ramp=10
... -set simulate.t_end=4000      # does the agreement hold?
```

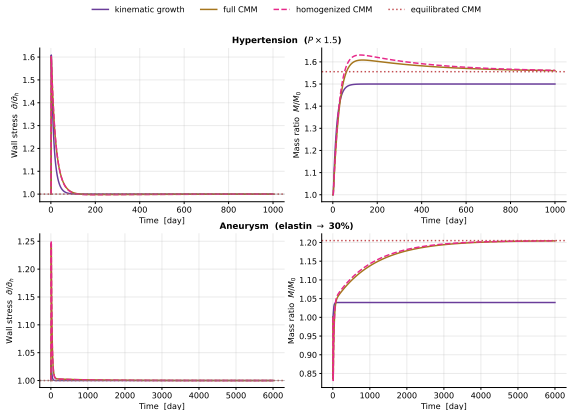
## Your turn: Homogenized vs. full

**Question.** Is the difference between the two curves visible by eye? By what factor is the homogenized model faster and does the agreement survive a harsher, longer insult?

**Hints.** Read the printed mass-ratio gap and the two runtimes. The homogenized model should track the full one closely wherever the loading is slow and gradual.

## ✓ Answer: Homogenized vs. full

The homogenized curve overlays the full CMM closely at a fraction of the runtime ( $O(N)$  vs.  $O(N^2)$ ). **Learning:** it reproduces a **similar time trajectory** to the full model, cheaply.



# Equilibrated CMM

# Directly Solve for the End State

Set every rate to zero and solve the **algebraic** equilibrium, one root-find instead of thousands of time steps.

**3D** the 3D equilibrium conditions

$$\underbrace{\Upsilon^\alpha = 1 \Rightarrow \bar{\sigma}^* = \bar{\sigma}_h}_{\text{(A) growth balance}} \quad \underbrace{\mathbf{F}_n^\alpha = \text{const} \Rightarrow \sigma^\alpha = \sigma_h^\alpha}_{\text{(B) remodeling completed}} \quad \underbrace{\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}}_{\text{(C) mechanical equilib.}}$$

**0D** scalar conditions at the evolved geometry

$$\bar{\sigma}^* = \bar{\sigma}_h, \quad \lambda_e^\alpha = G^\alpha, \quad \sigma_\theta = \frac{Pa}{h}.$$

# Collapse to One Scalar Equation

Let the turnover constituents grow in fixed proportion (exact when they share gain and turnover). The 3D conditions (A)–(C) collapse to a *single* equation for the evolved stretch  $\lambda^*$ , consistent with the same  $\lambda$ ,  $G^\alpha$ ,  $\phi^\alpha$  used throughout:

**0D** scalar equilibrium equation for  $\lambda^*$

$$\phi^e \xi + \xi \frac{\bar{\sigma}_h - \sigma^e(G^e \lambda^*)}{\sigma_{\text{turn}} - \sigma_h^e} - \gamma (\lambda^*)^2 = 0 \quad (21)$$

with the Laplace exponent  $n = 2$ ,  $\gamma$  the sustained load factor,  $\xi$  the surviving elastin fraction,  $\phi^e$  the elastin mass fraction,  $\sigma_h^e = \sigma^e(G^e)$  the elastin homeostatic stress, and  $\sigma_{\text{turn}}$  the homeostatic stress carried by the turnover constituents (collagen and smooth muscle). Once  $\lambda^*$  is known, the evolved mass ratio is  $M/M_0 = \gamma(\lambda^*)^2$ .

This equation *is* the fixed point of the homogenized ODEs ((19)–(20) with  $d/dt = 0$ ), so its root coincides with the long-time limit of the transient theory.

## The Punchline: Existence Sets the Capacity Limit

Equation (21) may have **no physical root**. Then no adapted state exists and the transient theories dilate without bound. So the solver is the cleanest **existence test**: no root  $\Rightarrow$  the wall lacks the *capacity* to adapt. Existence is *necessary* for stability; whether the transient actually *reaches* an equilibrium that exists is a separate, gain-dependent question (dynamic stability). Near the boundary the transient slows critically, which is exactly why the instant algebraic test is so useful.



## ⚙️ Setup: Example (the equilibrated stability test, ex04)

**Example.** Skip the transient: solve the single algebraic equation (21) for the *final* adapted stretch  $\lambda^*$ . A physical root means the tissue adapts; *no root* means it dilates without bound. Scanning elastin loss then traces the stability boundary.

### Run it.

```
uv run python run.py configs/ex04_equilibrated.yaml
```

Here `study: equilibrium` solves once for the given insult (and prints a transient cross-check: the homogenized plateau lands on  $\lambda^*$ ), then the `scan: block` sweeps `elastin_surviving` and `report_critical: elastin` prints the critical surviving fraction  $\xi_{\text{crit}}$ .

**Parametric study.** Move the boundary by changing the wall's load-bearing *capacity*, then re-read  $\xi_{\text{crit}}$ :

```
... -set model.constituents.collagen.law.c1=600
... -set model.constituents.collagen.G=1.15
```

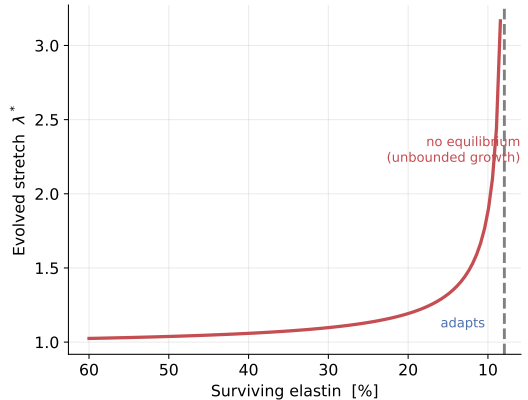
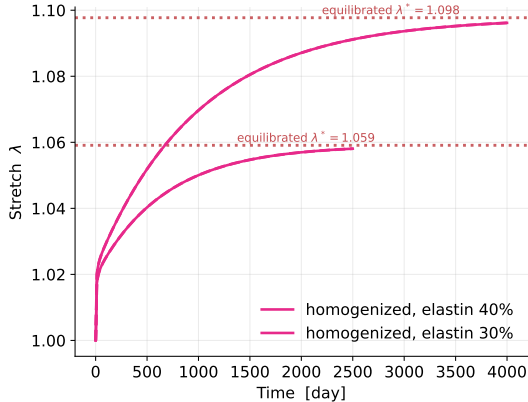
(Raising the gain leaves  $\xi_{\text{crit}}$  unchanged: gain sets how *fast* the transient adapts, not *whether* an equilibrium exists.)

## Your turn: Equilibrated

**Question.** Does the algebraic  $\lambda^*$  land on the transient plateau where a root exists? What is the critical elastin loss beyond which *no* equilibrium exists at all?

## ✓ Answer: Equilibrated

Where a root exists, the equilibrated solve lands exactly on the transient end-state. Below  $\xi_{\text{crit}}$  there is *no* root: the transient dilates without bound.

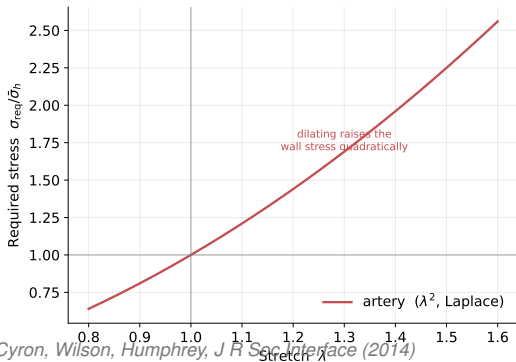


# Stability

# Where Instability Comes From: The Laplace Exponent

In a pressurised tube the wall stress is  $\bar{\sigma} = \frac{Pa}{h}$  with radius  $a \propto \lambda$  and thickness  $h \propto M$ , so the required (Laplace) stress is *quadratic* in the stretch,  $\bar{\sigma}_{\text{req}} \propto \gamma \lambda^2 / M$ . The homeostatic state is stable only if adding mass *lowers* the stress it must carry:

$$\left. \frac{d\bar{\sigma}}{d\theta} \right|_{\theta^*} < 0 \iff \text{tissue adapts} \quad (22)$$



Growth thickens the wall ( $h \propto M$ ), which lowers stress; but dilation raises it as  $\lambda^2$  (Laplace). While enough **elastin** survives, its stiff recoil keeps  $d\bar{\sigma}/d\theta < 0$  and the artery **adapts**. Lose enough elastin and the sign flips,  $d\bar{\sigma}/d\theta > 0$ : **runaway**. It is the quadratic Laplace exponent that makes this possible.

## Same Tissue, Two Insults

Read stability two ways: **existence**: the equilibrated equation (21) has a root iff the tissue can adapt (no time integration); or **transient**: integrate and watch it plateau or climb. The same tissue can do either: mild elastin loss adapts, severe loss dilates without bound.

## ⚙️ Setup: Example (the stable/unstable map, ex05)

**Example.** A tissue *adapts* to an insult iff a mechanobiological equilibrium exists (21). Sweeping that instant test over (surviving elastin)  $\times$  (pressure) draws the stable/unstable map. Then change *one* tissue property and watch the unstable “aneurysm” region shrink.

### Run it.

```
uv run python run.py configs/ex05_stability.yaml
```

Here study: map builds the 2D grid over the two axes: via the adapts existence test and shades adapt (blue) vs. runaway (red); it also prints the fraction of the grid that adapts.

**Parametric study.** Enlarge the *adapts* region by giving the wall more load-bearing capacity, then compare the printed fraction:

```
... -set model.constituents.collagen.law.c1=600
... -set model.constituents.collagen.G=1.15
```

(Raising the gain does *not* move this map: existence is gain-independent. Gain governs *dynamic* stability instead, in the transient.)

## Your turn: Stability map

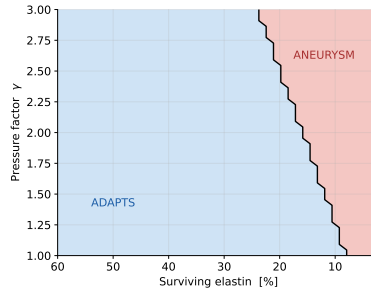
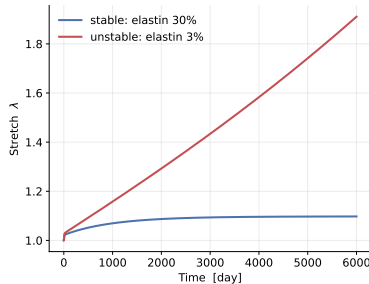
**Question.** Which capacity change (stiffer collagen or larger deposition stretch) enlarges the stable region the most, and why does raising the *gain* leave the map unchanged? Elastin cannot be regrown in adults: what does that imply for aneurysm?

**Hints.** Compare the printed “fraction that adapts”. Stiffness and deposition stretch move the existence boundary; the gain does not, it sets how fast (and whether) the transient reaches an equilibrium that exists.



## ✓ Answer: Stability map

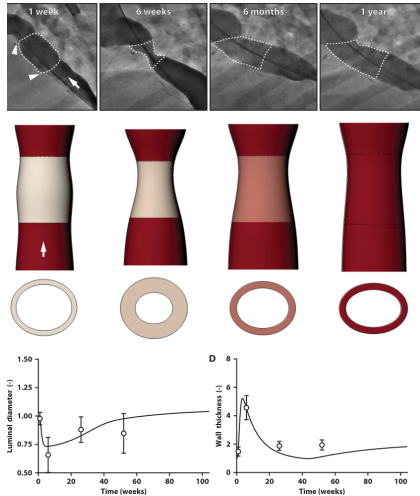
Stiffer collagen or a larger deposition stretch enlarge the *adapts* region: the wall carries more load, so it tolerates more elastin loss. The *gain* leaves this existence map unchanged (at equilibrium, production balances removal and the gain cancels); it instead sets *dynamic* stability, whether the transient actually reaches an equilibrium that exists (too weak adaptation runs away even when a root exists).



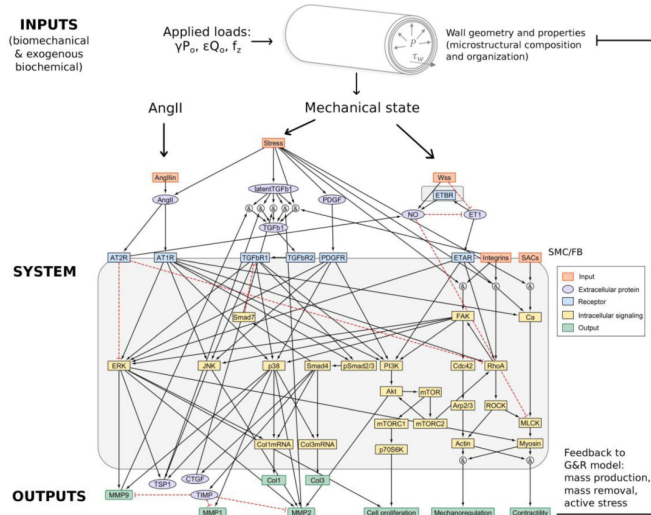
**Learning:** an equilibrium must *exist* (enough load-bearing capacity) **and** adaptation must be fast enough to *reach* it, existence is necessary, sufficient gain makes it so.

# Future Perspectives

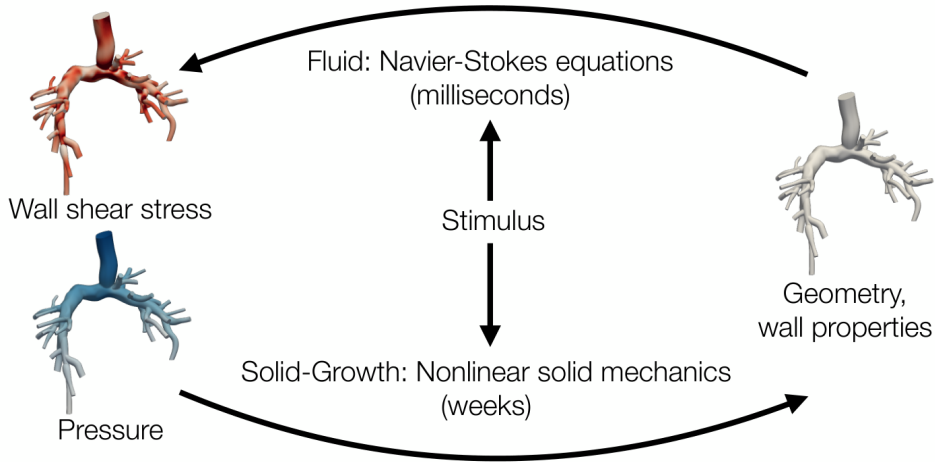
# Clinical Application: Reversible Graft Stenosis



# Systems-Biology Coupling: Signaling to Remodeling



# Fluid–Solid–Growth: Closing the Hemodynamic Loop

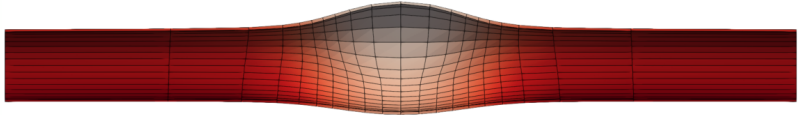


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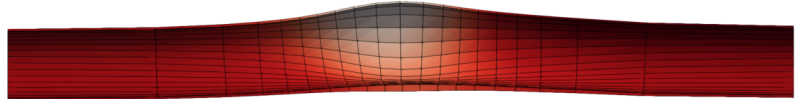
# FSGe: Long-Term Evolved Aneurysmal Equilibrium

**Solid with  $\tau_w \sim 1/r^3$**

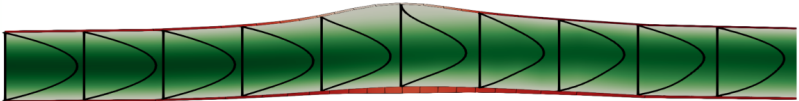
Wall shear  
stress



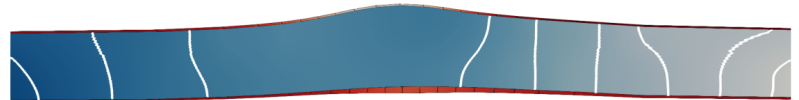
**Fluid-Solid-Growth  
interaction**



Velocity



Pressure



# Open-Source Implementations

# Under the Hood, What Each File Computes (1/2: the core)

The `gr` package is layered: a shared **continuum-mechanics core** (read, don't edit), then one file per G&R theory.

| File                       | Computes                                                                                                                          | Continuum mechanics / G&R                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <code>mechanics.py</code>  | strain energy $W(\lambda_e)$ , 2nd PK stress $S = dW/dE_e$ , Cauchy $\sigma = \lambda_e^2 S$ (neo-Hookean, Fung)                  | finite-strain <b>hyperelasticity</b> of each constituent, in reference measures |
| <code>parameters.py</code> | Model/Constituent $(\phi_0, G^\alpha, k_d, K_\sigma)$ ; set-point $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$ ; Insult            | <b>deposition prestretch</b> & homeo-static <b>set-points</b> ; the two insults |
| <code>geometry.py</code>   | required Laplace stress $\bar{\sigma}_h \propto \lambda^2/m$ ; solves $\bar{\sigma}(\lambda) = \sigma_{\text{req}}$ for $\lambda$ | <b>mechanical equilibrium</b> ; no root $\Rightarrow$ runaway                   |
| <code>history.py</code>    | Result: $t, \lambda, M/M_0, \bar{\sigma}, \dots$                                                                                  | shared time-history output                                                      |



# Under the Hood, What Each File Computes (2/2: the theories)

| File                             | Computes                                                                                                                           | G&R theory / role                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| kinematic_growth.py              | split $\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$ , scalar $\theta$ ; $\dot{\theta} = k_g \theta (\bar{\sigma} / \bar{\sigma}_h - 1)$ | Theory 1: kinematic growth                     |
| constrained_mixture.py           | heredity integrals over cohorts; mass balance + mixture stress ( $O(N^2)$ )                                                        | Theory 2: full CMM                             |
| homogenized_cmm.py               | two ODEs per constituent: $\dot{M}^\alpha, \dot{\lambda}_n^\alpha$ ( $O(N)$ )                                                      | Theory 3: homogenized CMM                      |
| equilibrated_cmm.py              | rates $\rightarrow 0$ : one algebraic solve for $\lambda^*$                                                                        | Theory 4: equilibrated CMM                     |
| stability.py                     | adapts (does a root exist?), critical_elastin_loss                                                                                 | stability = <b>existence</b> of an equilibrium |
| scenario.py, configs/<br>run.py, | build model/geometry/insult from YAML; run a study; sweep                                                                          | the exercise driver (input files)              |

Common thread: each theory changes only *how the supplied stress  $\bar{\sigma}(\lambda)$  is built*, the same constituent laws and equilibrium solve are reused, so every difference you saw is the G&R law alone.

# Research-Grade Open-Source Solvers

| Project             | What it does                                                                                      | Stack / group                                |
|---------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>svGrowth</b>     | constrained-mixture G&R of vessels (biaxial thin-wall); the cross-validation reference used above | Python; Stanford CBCL                        |
| <b>svFSGe</b>       | fast, strongly-coupled 3D fluid–solid–growth driver (the FSGe method)                             | Python + svFSI/svMultiPhysics; Stanford CBCL |
| <b>FEMBE_Plugin</b> | mechanobiologically equilibrated constrained-mixture material as a FEBio plugin                   | C++ / FEBio; Yale (Humphrey lab)             |
| <b>FEFSG_Plugin</b> | fluid–solid–growth coupling as a FEBio plugin (thoracic-aorta hypertension & aneurysm cases)      | C++ / FEBio; Yale (Humphrey lab)             |

[github.com/StanfordCBCL/svGrowth](https://github.com/StanfordCBCL/svGrowth)    [github.com/StanfordCBCL/svFSGe](https://github.com/StanfordCBCL/svFSGe)

[github.com/yale-humphrey-lab/FEMBE\\_Plugin](https://github.com/yale-humphrey-lab/FEMBE_Plugin)    [github.com/yale-humphrey-lab/FEFSG\\_Plugin](https://github.com/yale-humphrey-lab/FEFSG_Plugin)

# How the Four Theories Answer: The Takeaways

| Theory                     | Stability?       | What it says                                                                                                                             |
|----------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Kinematic growth           | <i>Imposed</i>   | Either reaches the <i>a-priori prescribed</i> homeostasis, or is unstable, nothing in between. Built in, not learned.                    |
| Constrained mixture (full) | <i>Predicted</i> | Adapts or dilates <b>depending on the insult</b> , the faithful reference.                                                               |
| Homogenized CMM            | <i>Predicted</i> | A <b>similar time trajectory</b> to the full model, at a fraction of the cost.                                                           |
| Equilibrated CMM           | <i>Instantly</i> | <b>Matches</b> the transient theories <i>if</i> an equilibrium exists; <i>no</i> root $\Rightarrow$ it does not match, unbounded growth. |

# Summary: What to Take Away

**The biology.** Living tissue continuously **turns over**; cells deposit new material *pre-stretched* at  $G^\alpha$ , chasing a homeostatic mechanical set-point  $\sigma_h$ .

**The four theories, one set-point, one question: does it adapt or run away?**

- **Kinematic growth:** prescribe  $F_g$ ; *imposes* the set-point. Outcome is binary: reach the prescribed homeostasis, or go unstable.
- **Constrained mixture:** track every cohort; stability is *predicted* and **depends on the insult** (mild  $\rightarrow$  adapt, severe  $\rightarrow$  runaway).
- **Homogenized:** one mean natural configuration per constituent; a **similar time trajectory** to the full model at  $O(N)$  not  $O(N^2)$ .
- **Equilibrated:** solve for the end state directly; **matches** the transient theories *iff* an equilibrium exists, and flags instability by finding no root.

**Where it goes.** These tools now drive **clinical prediction** (TEVGs), **systems-biology coupling**, and **fluid–solid–growth**.

# Key references I

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## Key references II

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- Schwarz, Pfaller, . . . , Marsden. A fluid–solid–growth solver for cardiovascular modeling. *Comput Methods Appl Mech Eng* (2023).
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- Humphrey, Eberth, Dye, Gleason. Fundamental role of axial stress in arterial mechanics. *J Biomech* (2009).
- Ambrosi, Ben Amar, Cyron, . . . , Humphrey, Kuhl. Growth and remodelling of living tissues. *J R Soc Interface* (2019).

# Key references III

## Code

- Seminar code (this deck's figures): `github.com/YaleCBL-Teaching/Growth-Remodeling`
- `github.com/StanfordCBCL/svGrowth`, `svFSGe`
- `github.com/yale-humphrey-lab/FEMBE_Plugin`, `FEFSG_Plugin`

**Figure sources.** Result figures reproduced with the `gr` package; literature figures cited per-slide.